

Same Counts, Different Message

AP Statistics · Topic 1.4 (CED 1.4) · B: 2026-09-14 · E: 2026-09-16 · TEACHER KEY

CED TETHER

- Enduring Understanding: UNC-1 Graphical representations and statistics allow us to identify and represent key features of data.
- Skills:
- **Skill 2.B** — Construct numerical or graphical representations of distributions.
- **Skill 2.A** — Describe data presented numerically or graphically.
- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- Learning Objectives and Essential Knowledge:
- LO UNC-1.C Represent categorical data graphically. [Skill 2.B]
- EK UNC-1.C.1 Bar charts (or bar graphs) are used to display frequencies (counts) or relative frequencies (proportions) for categorical data.
- EK UNC-1.C.2 The height or length of each bar in a bar graph corresponds to either the number or proportion of observations falling within each category.
- EK UNC-1.C.3 There are many additional ways to represent frequencies (counts) or relative frequencies (proportions) for categorical data.
- LO UNC-1.D Describe categorical data represented graphically. [Skill 2.A]
- EK UNC-1.D.1 Graphical representations of a categorical variable reveal information that can be used to justify claims about the data in context.
- LO UNC-1.E Compare multiple sets of categorical data. [Skill 2.D]
- EK UNC-1.E.1 Frequency tables, bar graphs, or other representations can be used to compare two or more data sets in terms of the same categorical variable.

Essential question: How can a graph use correct counts but still create a misleading comparison?

DOK-3 skill: 4.B · **FRQ pattern:** truncated-axis-which-display-is-honest-and-what-it-misleads

DOK rationale: Part (c) adjudicates two displays, connects the baseline to distortion, contrasts apparent and actual magnitudes, and defends a second display choice for a stated purpose.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Read both captions aloud, then collect first takes without reacting.
- Begin the video follow-along at minute 5.
- Return to the graphs at minute 33. Circulate and require students to connect the baseline to bar height.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Write a one-sentence first take that names the graph they trusted.
- Complete the video follow-along on categorical displays.

- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- If the counts are identical, what changed between the two graphs?
- From what vertical value are you measuring each visible bar in Graph A?
- How does the visible 12-to-1 comparison differ from the actual 52-to-41 comparison?
- For four close categories, what can your eyes compare more precisely: aligned lengths or wedge angles?

Adult role

Solo teacher. Circulate 5 pairs. Do not call a graph honest or misleading for students; ask them to name the baseline, quantify both comparisons, and connect the display choice to the reader's question.

[WARN] WATCH FOR

- Calling Graph A's counts incorrect instead of explaining how its 40 baseline distorts their visual sizes.
- Treating the visible heights 12 and 1 as the original category counts.
- Choosing a pie chart because the data can be percentages without considering whether close wedges are easy to compare.

SCORING PART (c) — E / P / I

Expected elements

- **selects-zero-baseline-graph** (required) — Selects Graph B and connects its zero baseline to bar height representing count
- **quantifies-visual-distortion** (required) — Uses visible heights 12 and 1 to explain the apparent 12-times ratio
- **quantifies-actual-difference** (required) — Contrasts that with 52 versus 41, 11 students, or about 27 percent
- **chooses-display-for-purpose** (required) — Defends bar or pie by what lengths versus angles or areas make easier to compare

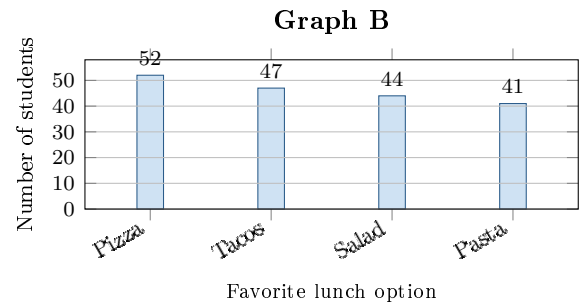
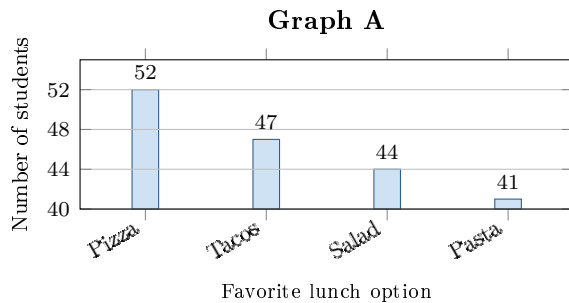
E	Chooses Graph B with the baseline mechanism, quantifies both apparent and actual differences, and defends a display for comparing close categories.
P	Makes the graph decision but omits the mechanism or one numerical contrast, or gives only a generic display reason.
I	Trusts Graph A's magnitude, treats 12 and 1 as counts, or gives no feature-based display choice.

Common mistakes

- Calling Graph A's counts false instead of explaining its truncated baseline
- Reading visible heights 12 and 1 as student counts
- Saying 52 is 11% more than 41 rather than dividing by 41

[STAR] TODAY'S PROBLEM — annotated

Suppose 184 students chose a favorite lunch option. A newsletter printed **Graph A** with the headline “**Pizza wins by a landslide.**” A council member redrew the data as **Graph B** and said, “It is basically a tie.”



FIRST TAKE (5 min)

Is pizza clearly the favorite? Name the graph that most influenced you.

Key: Not graded. Expect Graph A to attract votes because its Pizza bar dominates.

Rules callout “GRAPHING ONE CATEGORICAL VARIABLE (keep on the page)” is printed on the student sheet.

DOK 1 (a) Read the Pizza and Pasta counts. How many more students chose Pizza?

Key: Pizza has 52 students and Pasta 41, so Pizza has 11 more.

DOK 2 (b) Convert all four counts to whole-number percents of 184. Describe how close the categories are in context.

Key: Pizza 28%, Tacos 26%, Salad 24%, Pasta 22% (rounded). The largest and smallest shares differ by about 6 percentage points, and none reaches 30%.

DOK 3 (c) For “Is pizza clearly the favorite?” choose the more honest graph and name the design feature that settles it. Quantify Graph A’s visible Pizza-to-Pasta heights, then compare 52 with 41 using percent increase. Finally, choose a zero-baseline bar graph or pie chart for comparing the four categories and defend that choice.

Key: Graph B: its zero baseline makes full bar height proportional to count. Graph A starts at 40, so visible heights are 12 and 1; Pizza looks 12 times as tall. In fact, $52/41 = 1.27$: only 11 students, about 27% more (about a 6-point share lead). Pizza leads narrowly—neither a landslide nor a tie. Use the zero-baseline bar graph because aligned lengths reveal close differences; a pie chart emphasizes shares of a whole but its similar wedges are harder to compare.

EXIT

One thing the video changed about my first take: _____